

1. The CNN Architecture

The model uses a custom **Multi-Scale Spatial Encoder** ([class MultiScaleSpatialEncoder](#)). It is not a standard off-the-shelf CNN like ResNet or VGG, but rather a lightweight, multi-branch architecture designed for efficiency and feature diversity.

Key Characteristics:

- Stem:** A standard convolutional block that downsamples the input image (64x64) using [Conv2d](#) + [BatchNorm](#) + [ReLU](#) followed by [MaxPool2d](#).
- Multi-Scale Branches:** It splits the feature map into three parallel branches to capture features at different scales:
 - Scale 1:** Processes features at the current resolution.
 - Scale 2:** Downsamples further using [Depthwise Separable Convolutions](#) and [MaxPool2d](#).
 - Scale 3:** Downsamples even further for high-level semantic features.
- Depthwise Separable Convolutions:** Instead of standard convolutions, it uses [depthwise_sep_conv](#) (a depthwise convolution followed by a pointwise 1x1 convolution). This significantly reduces the number of parameters and computational cost.
- CBAM Attention:** Each branch incorporates **CBAM (Convolutional Block Attention Module)**, which consists of:
 - SE Block (Squeeze-and-Excitation):** Channel-wise attention.
 - Spatial Attention:** Spatial-wise attention using average and max pooling.
- Attention Pooling:** Instead of global average pooling, it uses a learned [AttentionPool2d](#) to weight spatial locations before flattening.
- Fusion:** The outputs from the three scales are concatenated and projected to the transformer dimension ([D_MODEL](#)) via a Linear layer.

2. The Transformer Architecture

The model uses a **Standard Encoder-Only Transformer** ([nn.TransformerEncoder](#)) adapted for temporal sequence processing.

Key Characteristics:

- Type:** It is an **Encoder-only** architecture (similar to BERT's encoder stack, but without self-attention masking since it processes causal time series in chunks). It does not use a Decoder (like GPT or Vision Transformers for generation).
- Input:** It takes a sequence of frame embeddings (output from the CNN) of shape ([Batch](#), [Sequence_Length](#), [D_Model](#)).
- Positional Encoding:** Uses **Sinusoidal Positional Encoding** (fixed, not learned) to inject temporal order information into the sequence.
- Transformer Layers:**
 - Uses PyTorch's built-in [nn.TransformerEncoderLayer](#).
- Norm First:** Uses [norm_first=True](#) (Pre-LN), which is more stable for training deep transformers.

- Activation:** Uses **GELU** activation instead of ReLU.
- Hyperparameters:**
 - [d_model](#): 128
 - [nhead](#): 8
 - [dim_feedforward](#): 512
 - [num_layers](#): 4
 - [dropout](#): 0.1
- Post-Transformer Modules:**
 - Spectral Attention 1D:** A custom module that applies attention based on the frequency domain magnitude of the temporal features.
 - Temporal Convolutional Residual Block:** A lightweight 1D CNN block (Depthwise Conv -> Pointwise Conv -> BatchNorm -> GELU -> Conv) added as a residual connection after the Transformer. This helps capture local temporal dependencies that the Transformer might miss.

Summary of the Hybrid Model ("RPPGTransformer")

- 1.**Dual-Stream Input:** The model processes two streams of video frames:
 - Motion Stream:** Frame differences ([diff_frames](#)).
 - Appearance Stream:** Raw frames ([raw_frames](#)).
- 2.**CNN Feature Extraction:** Each stream passes through its own instance of the [MultiScaleSpatialEncoder](#) (CNN) to extract spatial features per frame.
- 3.**Fusion:** The features from both streams are concatenated and fused via a Linear layer.
- 4.**Temporal Modeling:** The fused sequence passes through the **Transformer Encoder** to model long-range temporal dependencies.
- 5.**Refinement:** The output is refined by Spectral Attention and a Temporal Convolutional residual block.
- 6.**Regression Head:** Finally, a simple Linear head maps the temporal features to the rPPG signal (BVP waveform).

This architecture is a **Hybrid CNN-Transformer**, leveraging the CNN's strength in spatial feature extraction and the Transformer's strength in modeling long-range temporal dynamics, specifically tailored for remote photoplethysmography (rPPG).